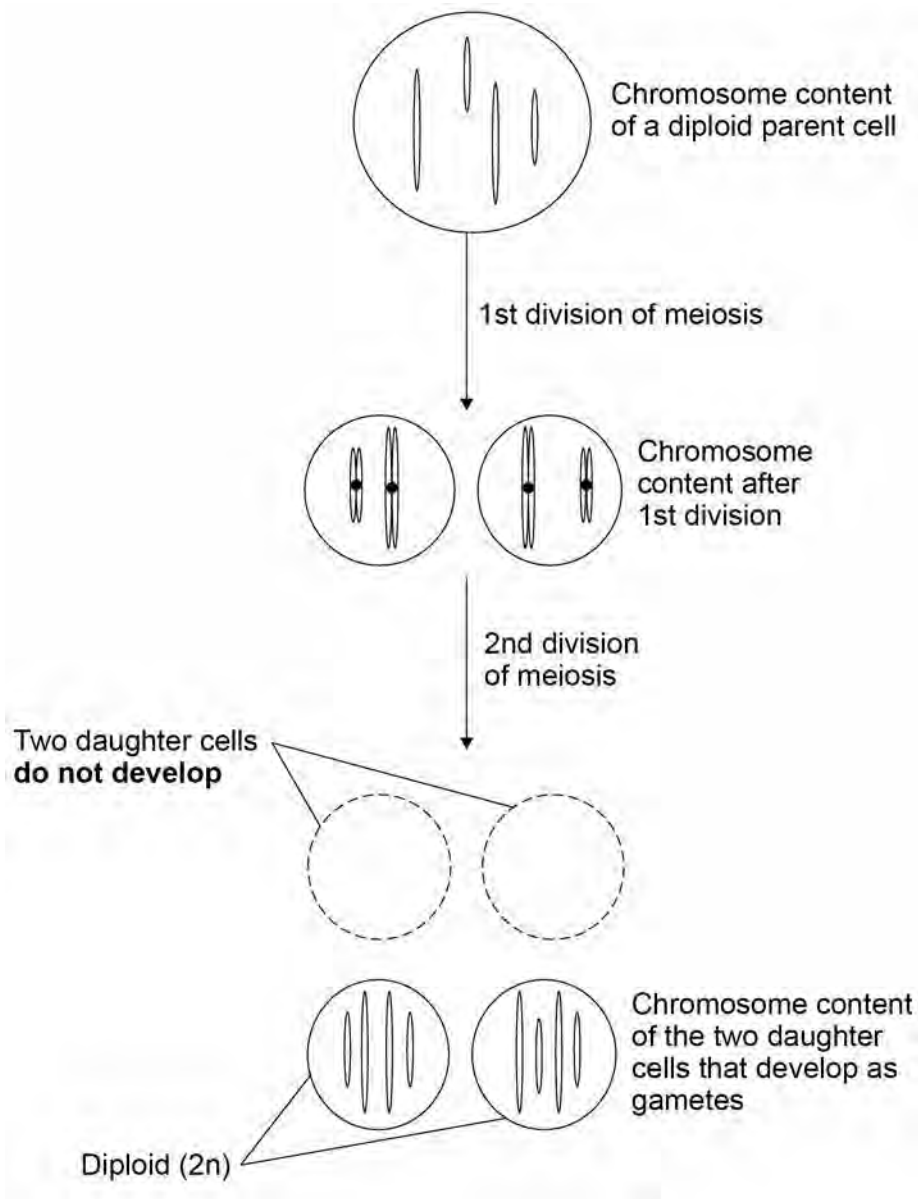


Figure 6 shows a faulty form of meiosis that can occur in some plants.

Figure 6

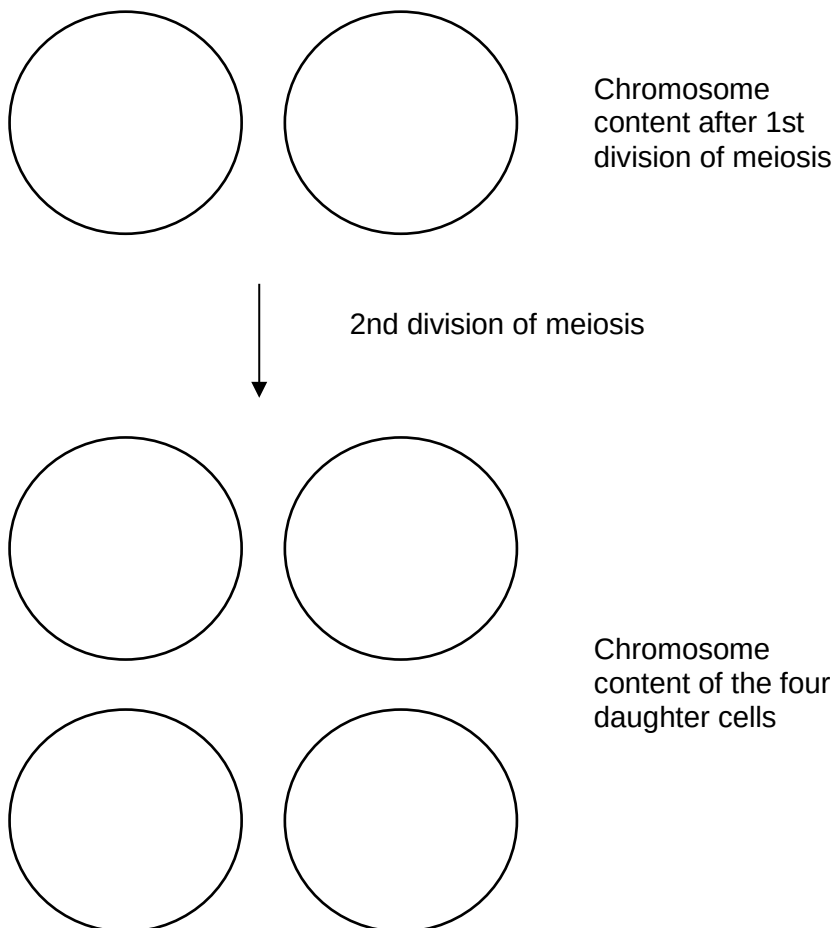


0 6 . 1

Complete **Figure 7** to show the chromosome content of the cells that would result from a normal meiotic division of the diploid parent cell shown in **Figure 6**.

[2 marks]

Figure 7



Question 6 continues on the next page

Turn over ►

0 6 . 2

If two diploid ($2n$) gametes fuse at fertilisation, it can result in the growth of a tetraploid plant which has 4 copies of each chromosome.

Red clover is a plant grown to produce cattle feed. Tetraploid red clover plants produce a higher yield than diploid red clover plants.

Whether a red clover plant produces $2n$ gametes is genetically controlled.

Scientists investigated the possibility of breeding red clover plants that only produced $2n$ gametes.

- In breeding cycle 0, they grew red clover plants and identified plants that produced $2n$ gametes.
- In breeding cycle 1, they used the plants producing $2n$ gametes to produce offspring.
- In breeding cycles 2 and 3, they identified plants producing $2n$ gametes and used these to produce offspring.

Their results are shown in **Table 3**.

Table 3

Breeding cycle	Observed		Expected	
	Number of plants that did not produce $2n$ gametes	Number of plants that did produce $2n$ gametes	Number of plants that did not produce $2n$ gametes	Number of plants that did produce $2n$ gametes
0	50	4	50	4
1	14	42		
2	2	44		
3	0	56		

The scientists used the following null hypothesis.

'The proportion of plants that produce $2n$ gametes will not change from one breeding cycle to the next.'

Complete **Table 3** to show the **expected number** of plants that **did not** produce $2n$ gametes and the expected number of plants that **did** produce $2n$ gametes after 1 cycle.

Give each answer to the nearest whole number.

[2 marks]

0 6 . 3

The scientists tested their null hypothesis using the chi-squared statistical test. After 1 cycle their calculated chi-squared value was 350
The critical value at $P=0.05$ is 3.841

What does this result suggest about the difference between the observed and expected results and what can the scientists therefore conclude?

[2 marks]

0 6 . 4

Use your knowledge of directional selection to explain the results shown in **Table 3**.

[3 marks]

[Extra space]

[

